

8086 Memory Segmentation – Easy Explanation

■ The Trick:

8086 processor can access 1 MB memory using 20-bit addresses, but registers are only 16-bit. So, it divides memory into **segments** (blocks of 64 KB each).

Every memory address is represented using:

$$\text{Physical Address} = (\text{Segment} \times 10\text{h}) + \text{Offset}$$

■ Example:

If Segment = 2000h and Offset = 0000h

$$\rightarrow (2000\text{h} \times 10\text{h}) + 0000\text{h} = 20000\text{h}$$

If Offset = FFFFh

$$\rightarrow (2000\text{h} \times 10\text{h}) + \text{FFFFh} = 2\text{FFFFh}$$

So the segment covers **20000h to 2FFFFh** (total 64 KB).

Part	Meaning	Bits	Example
Segment	Starting block of memory	16-bit	2000h
Offset	Distance inside the block	16-bit	0010h
Physical Address	Actual 20-bit address	20-bit	$(2000\text{h} \times 10\text{h}) + 0010\text{h} = 20010\text{h}$

■ Easy Way to Remember:

Think of memory like a city divided into 64 KB areas (segments).

Each house in a segment is found using an offset.

Segment = Area number, Offset = House number, Physical Address = Full address.